

13 Introduction to Catalysis

Why are we learning this?

Catalysis is responsible for many of the most important chemical technologies in modern society. Catalysts are used to produce:

- Fuels
- Fertilizers
- Polymers
- Pharmaceuticals
- Environmental remediation technologies

Most industrial chemical reactions rely on catalysts. In this course, we are interested in catalysts because catalysts influence reaction rates, reactor design, reactor performance, and process economics. By the end of today's lesson, you should understand:

- What a catalyst is.
- What catalysts do and do not do.
- How catalysts influence reaction rates.
- The difference between homogeneous and heterogeneous catalysis.
- Why catalysts lower activation barriers.
- Why catalytic reactions often proceed through different mechanisms than uncatalyzed reactions.

Course Roadmap

Introduction to CRE

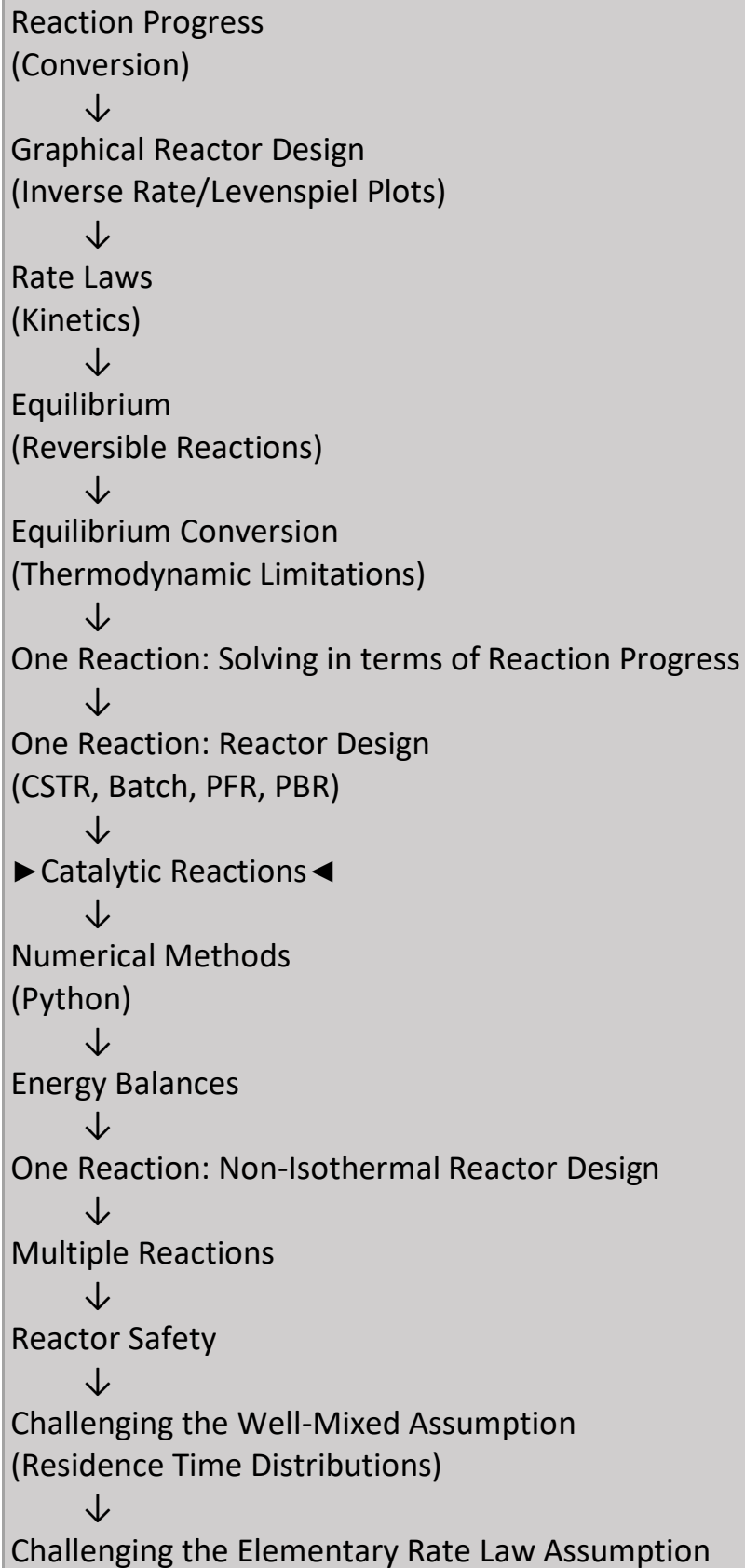


General Balance Equation



Common Reactor Models
(CSTR, Batch, PFR)





(Reaction Mechanisms)

Big Idea

Catalysts do not change where equilibrium lies. Catalysts change how quickly equilibrium is reached. This is one of the most important ideas in catalysis.

A catalyst is

Catalysts provide lower energy routes from reactants to products without being consumed.

The catalyst influences

What Is a Catalyst?

A catalyst is a substance that increases the rate of a chemical reaction while being regenerated at the end of the reaction cycle. A catalyst participates in the reaction mechanism, but it is not permanently consumed. This is why relatively small amounts of catalyst can influence large quantities of reacting material.

What Does a Catalyst Actually Do?

Catalysts do not create new products. Catalysts do not change reaction stoichiometry. Catalysts provide an alternative reaction pathway that requires less activation energy. Lower activation energy generally leads to faster reaction rates.

Catalysts Affect Kinetics, Not Equilibrium

Catalysts influence:

Reaction Rate

Catalysts do **not** influence:

Equilibrium Constant

Equilibrium Conversion

Reaction Thermodynamics (e.g., Heat of Reaction)

A catalyst helps a system reach equilibrium faster, but it does not change where equilibrium occurs.

Types of catalysts:

- Homogeneous catalysts:

Examples: Ion/acid catalysts, organometallic catalysts, enzymes

Example: Cl atoms catalyze the decomposition of ozone O_3 :

In the absence of Cl: $O + O_3 \rightarrow 2O_2$, activation energy $E = 20 \text{ kJ/mol}$

In the presence of Cl:

Energy diagram:

This
drawing is
not to
scale; it's
just for
illustrative
purposes!

The catalyst provides

Reading an Energy Diagram

The highest point on the energy diagram corresponds to the transition state. The vertical distance between reactants and the transition state is the activation energy. A catalyst lowers this barrier by providing an alternative pathway.

Lower Activation Energy



Larger Rate Constant



Faster Reaction

What Can We Learn from the Ozone Example?

The uncatalyzed and catalyzed reactions produce the same overall products. The difference is that the catalyst creates a new reaction pathway. Because the catalytic pathway has a lower activation barrier, the reaction proceeds more rapidly. This example illustrates the central role of catalysts in changing kinetics without changing overall chemistry.

Two Major Classes of Catalysts

Homogeneous Catalysts: Catalyst and reactants exist in the same phase.

Examples:

- Acid catalysts
- Organometallic catalysts
- Enzymes

Heterogeneous Catalysts: Catalyst and reactants exist in different phases.

Examples:

- Supported metal catalysts
- Zeolites
- Ziegler–Natta catalysts

Most industrial reactor systems use heterogeneous catalysts.

- Heterogeneous catalysts:

Examples: Supported metal catalysts, zeolite (molecular sieves), Ziegler-Natta catalyst (TiCl_4 anchored to MgCl_2)

Example: CO oxidation on $\text{Pt}/\text{Al}_2\text{O}_3$

* = catalyst
"sites"
*'ed species
mean those
species are
bonded to/
adsorbed to
the catalyst.

O₂ doesn't dissociate in the gas phase, but it readily adsorbs to the catalyst and dissociates on the catalyst. The catalyst provides a lower energy route from reactants to products.

Why Surfaces Matter in Heterogeneous Catalysis

Many heterogeneous catalysts work because molecules can adsorb onto catalytic surfaces. Adsorption can:

- Bring reactants together.
- Orient reactants favorably.
- Weaken chemical bonds.
- Create lower-energy reaction pathways.

The catalyst surface becomes an active participant in the reaction mechanism.

Interpreting Catalytic Reaction Mechanisms

In catalytic mechanisms * represents an available catalytic site.

Species written as A* are adsorbed onto the catalyst surface.

The catalyst surface is often treated as a reactant in the steps in the catalytic mechanism because catalytic sites participate directly in the mechanism. However, the overall mechanism must regenerate all catalyst sites used during the reaction.

Summary and Key Takeaways

Today we introduced the basic concepts of catalysis.

Key Ideas

- ✓ Catalysts increase reaction rates without being permanently consumed.
- ✓ Catalysts provide alternative reaction pathways with lower activation energies.
- ✓ Catalysts affect kinetics, not thermodynamics.
- ✓ Homogeneous and heterogeneous catalysts operate in different phases.
- ✓ Many heterogeneous catalysts function through adsorption of reactants onto catalytic surfaces.

✓ Catalytic mechanisms often differ substantially from uncatalyzed mechanisms.

Most Important Idea

A catalyst does not change where a reaction goes. A catalyst changes how quickly it gets there. Understanding this idea is the foundation for catalytic reactors, catalytic rate laws, and catalytic reaction mechanisms.